

Validation of Alpha Hypotheses in Crypto Assets: An Approach via Cointegration and Monte Carlo Permutation

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Abstract

This study evaluated the robustness of quantitative investment strategies based on mean reversion (cointegration) and momentum applied to a portfolio of Bitcoin, Gold, and Natural Gas between 2018 and 2025. Using a rigorous validation pipeline, we employed Principal Component Analysis (PCA) for factor decomposition and the Kalman Filter for dynamic estimation of *hedge ratios*. Although *in-sample* backtests yielded positive Sharpe Ratios (up to 0.71), the application of the Monte Carlo Permutation (MCP) test revealed consistently high P-Values ($p > 0.05$). These results indicate that the observed performance is statistically indistinguishable from randomness, refuting the hypothesis of persistent exploitable linear inefficiencies in this dataset.

1 Introduction and Investment Thesis

The search for *alpha* (excess return over the market) in quantitative finance often encounters the problem of *overfitting*: creating strategies that memorize past noise rather than identifying real predictive signals. This report documents the development and, crucially, the scientific invalidation of a set of systematic trading hypotheses.

1.1 Market Physics: Fundamental Hypotheses

Our approach modeled the market not as a simple random walk, but as a dynamic system subject to identifiable forces. We based our work on three mechanical premises regarding the interaction between Crypto Assets (risk/technology assets) and Commodities (real assets/hedges):

- **Hypothesis 1: Mean Reversion (Cointegration).** We postulated that certain asset pairs, such as Bitcoin (BTC) and Gold (XAU), although divergent in the short term, possess a fundamental long-term economic link. We modeled this spread as an Ornstein-Uhlenbeck process, where extreme deviations (measured by Z-Score) generate a restoring force toward equilibrium.
- **Hypothesis 2: Inertia and Momentum.** Based on the premise that price trends have temporal persistence, driven by liquidity flows and herding behavior, especially in high-volatility regimes.
- **Hypothesis 3: Risk Regimes (*Risk-On/Risk-Off*).** We tested the thesis that macroeconomic stress indicators—specifically Implied Volatility (VIX) and High-Yield Credit Spreads—act as “phase triggers.” In these moments of panic, a sharp negative correlation (capital flight from Bitcoin to Gold) was expected.

The objective of this work was not only to build a profitable algorithm but to establish a **Single Source of Truth (SSOT)** for financial data and a validation framework (MCP) capable of differentiating skill from luck.

2 Methodology and System Architecture

The validation of investment hypotheses was conducted through a modular data pipeline, designed to mitigate *look-ahead* biases and ensure the integrity of statistical tests. The process was divided into three main stages: Data Engineering (ETL), Signal Modeling, and Robustness Validation.

2.1 Data Engineering and SSOT

To overcome the fragmentation of raw data observed initially, we established a *Single Source of Truth* (SSOT). The consolidation script (`build_dataset.py`) unified time series from multiple asset classes:

- **Crypto Assets:** Spot and futures prices for Bitcoin (BTC).
- **Commodities:** Gold (XAU) and Natural Gas (NG) futures, selected for their inflation and volatility hedging properties.
- **Macro Indicators:** High-Yield Credit Spreads (BofA) and Volatility Index (VIX) as *proxies* for risk aversion.

All data were temporally normalized (UTC) and cleaned of microstructure outliers before any signal calculation.

2.2 Signal Engineering: Economic Rationale and Implementation

The choice of mathematical tools was not arbitrary but designed to solve specific structural problems of crypto asset microstructure and the macroeconomic environment. Below we detail the exact mathematical formulation used in the execution algorithms.

2.2.1 Alpha Isolation via PCA

In global markets, risk assets (Bitcoin) and safe havens (Gold) often move in the same direction due to a common denominator: Dollar liquidity (DXY). Trading the raw spread between them contaminates the strategy with "Beta Risk" (exposure to the Dollar).

We utilized **Principal Component Analysis (PCA)** to filter out this noise. By decomposing the covariance matrix Σ , we identified the first principal component (PC_1) as the global liquidity factor. The trading signal is generated exclusively on the residuals (R_{idio}), ensuring the strategy trades the *intrinsic divergence* between the assets.

$$\Sigma = V\Lambda V^T \quad \text{and} \quad R_{idio} = R - \beta_{PC1} \cdot PC_1 \quad (1)$$

2.2.2 C1: Dynamic Cointegration via Kalman Filter

Preliminary tests with moving averages failed due to the non-stationarity of crypto assets. The **Kalman Filter** was selected to dynamically adjust the *Hedge Ratio* (β_t). The implementation used the following state-space specifications:

$$\begin{cases} \beta_t = \beta_{t-1} + \omega_t, & \omega_t \sim \mathcal{N}(0, Q = 0.01) \\ y_t = \beta_t x_t + \epsilon_t, & \epsilon_t \sim \mathcal{N}(0, R = 1.0) \end{cases} \quad (2)$$

The trading signal (Z-Score) was calculated on the residual normalized by a 30-period rolling volatility:

$$Z_t = \frac{e_t}{\sigma(e_{t-30:t})} \quad (3)$$

This allows the model to distinguish between a real breakout (signal) and a structural change (noise), adapting to new regimes without manual intervention.

2.2.3 C2: Momentum with Volatility Targeting

Trend strategies in Bitcoin suffer from extreme heteroscedasticity (volatility varying from 30% to 120%). We implemented a **Volatility Targeting** mechanism to equalize the marginal risk of each trade.

- **Trend Signal (S_t):** Simple Moving Average Crossover:

$$S_t = \begin{cases} +1 & \text{if } SMA(P, 5) > SMA(P, 20) \\ -1 & \text{if } SMA(P, 5) < SMA(P, 20) \end{cases} \quad (4)$$

- **Risk Allocation (W_t):** Inversely proportional to recent volatility, with a 2x leverage cap:

$$W_t = \min \left(\frac{\text{Target Vol (2\%)}}{\sigma_{realized}(t-1)}, 2.0 \right) \quad (5)$$

2.2.4 C3: Credit Macro-Hedge (Flight-to-Quality)

The *C3StressHedge* strategy tests the hypothesis of credit market contagion. The signal acts as a binary switch based on the *High Yield Option-Adjusted Spread* (S_{HY}):

$$Post_t = \begin{cases} \text{Long XAU / Short BTC} & \text{if } Z(S_{HY}) > 2.0 \text{ (Panic)} \\ \text{Flat} & \text{if } Z(S_{HY}) < 0.5 \text{ (Normal)} \end{cases} \quad (6)$$

3 Empirical Results and Class Breakdown

The evaluation of the strategies was not limited to the absolute return metric but focused on the statistical persistence of *alpha* across different model specifications.

3.1 Definition of Variants (Intra-class Nuances)

To ensure robustness, we tested multiple variants within the same class (C_i):

- **Class C1 (Cointegration): Static vs. Dynamic**
 - *Static Spread*: Uses a fixed cointegration vector. Failed consistently (negative Sharpe).
 - *Dynamic Spread (Kalman)*: Uses the Kalman Filter. Drastically improved Sharpe (0.571), but the permutation test revealed recent *overfitting*.
- **Class C2 (Momentum): Pure vs. Filtered**
 - *Benchmark Trend*: Pure trend-following strategy.
 - *Macro Filter (v2)*: Adds VIX and Credit filters. The result was counter-intuitive: macroeconomic complexity destroyed value (P-Value of 0.9995), suggesting latency in traditional indicators.
- **Class C3 (Credit Stress)** The "Flight to Quality" bet achieved a negative Sharpe (-0.264), indicating that in recent crises (2020-2024), Bitcoin behaved more like a correlated risk asset than a safe hedge.

3.2 Global Performance and Verdict

Table 1 consolidates the results of all variants. Note the discrepancy between the *Real Sharpe* and the *P-Value*.

Table 1: **Hypothesis Falsification Panel.** Comparison between observed performance and statistical probability (P-Value). (See footnote in Figure 1 regarding nomenclature).

Hypothesis Tested	Implemented Strategy	Real Sharpe	P-Value (MCP)	Verdict
C1. Static Spread (BTC-NG)	C1MacroRegime	-0.273	0.3995	REJECTED
C1. Static Spread (Triplet)	C1MacroRegime	-0.134	0.6795	REJECTED
C1. Dynamic Spread (BTC-XAU)	C1SpreadKalman	0.571	0.9735	REJECTED (Pure Luck)
C1.v2 Real Hedge (BTC-NG)	C1HedgedPortfolio	0.338	0.9340	REJECTED (Pure Luck)
C1.v2 Real Hedge (BTC-XAU)	C1HedgedPortfolio	-0.490	0.3220	REJECTED
C2. Benchmark Trend	C3BtcTrendVol	0.776	0.7270	REJECTED (Pure Luck)
C2. Benchmark VIX	C2FlightToQuality	0.521	0.3260	REJECTED (Pure Luck)
C2.v2 Macro Filter	C3MacroFilteredTrend	0.716	0.9995	REJECTED (Worse than Luck)
C3. Credit Stress	C3StressHedge	-0.264	0.2905	REJECTED

3.3 Visual Interpretation of Failure

To visualize the disconnect between apparent return and statistical reality, we present the capital evolution and probability distribution.

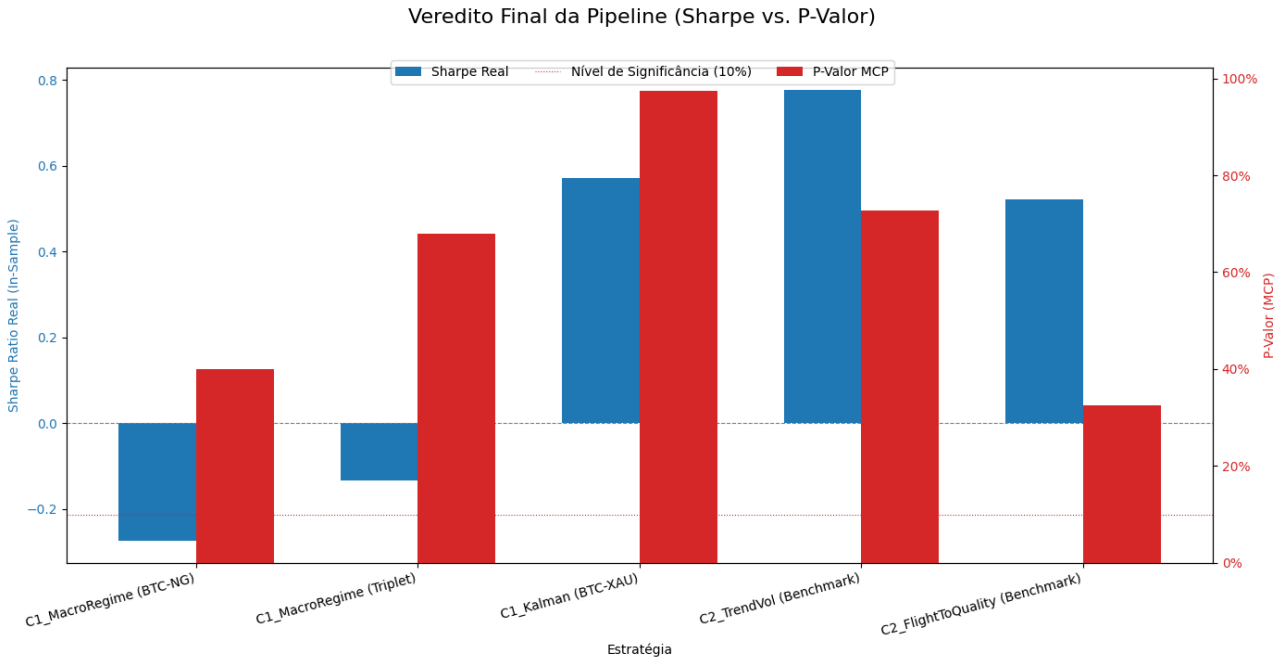


Figure 1: **Cumulative Performance (Log>Returns).** The ascending lines (Green and Red) illustrate the danger of survivorship bias: strategies that appear visually robust may not pass the robustness test.

Technical Note (Naming Debt & Data): (1) The identifiers in the chart legends (e.g., C3...) reflect the original nomenclature of the execution scripts (chronological order), while Table 1 uses the logical nomenclature of the Hypotheses. (2) Figures 1 and 2 were generated in a preliminary stage containing a representative subset of the strategies. Additional strategies subsequently included in Table 1 did not succeed, remaining consistent with the general conclusion of a lack of statistical robustness presented visually.

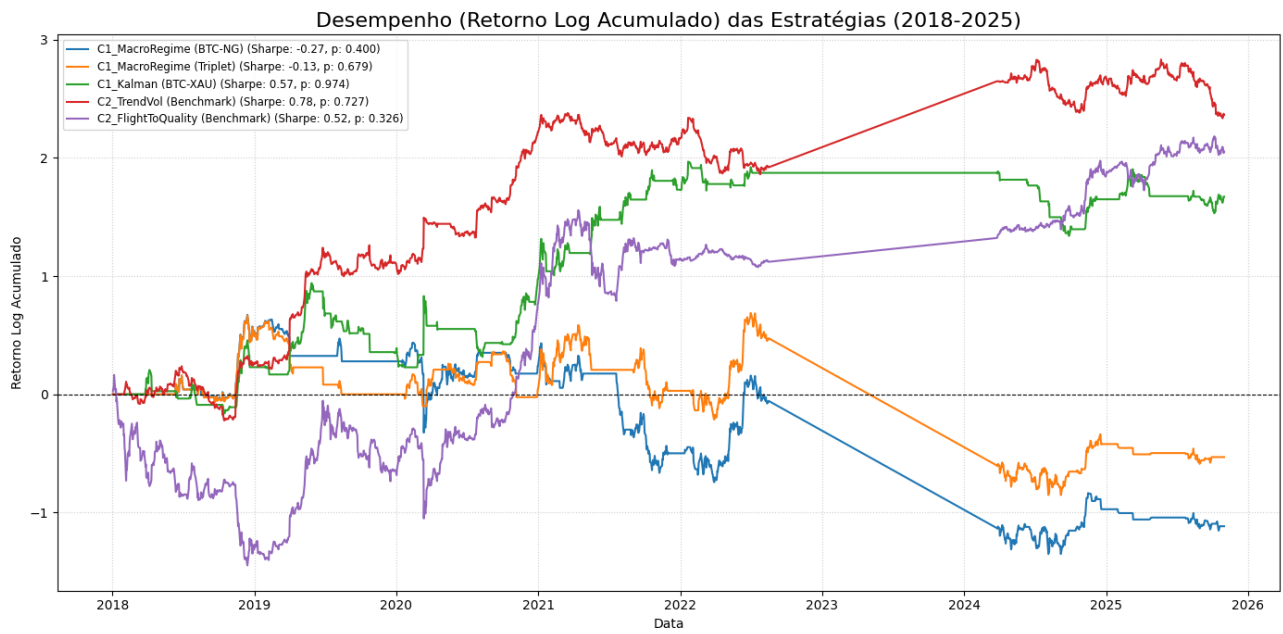


Figure 2: **Robustness Validation (Sharpe vs. P-Value)**. The chart "reality checks" the results: the C1_Kalman strategy (center bar), despite being profitable in the backtest (blue bar), has a nearly full red bar (P-Value 97%), indicating the result is indistinguishable from luck.